FUNNEL ANALYSIS

Portfolio Project

Multi-company funnel analysis across three real-world datasets representing three distinct funnel archetypes

CONVERSION	RETENTION	FULFILMENT

Dataset	Source	Funnel Type
Google Merchandise Store	GA4 Event Data - Nov 2020 to Jan 2021	Conversion
Instacart Grocery Delivery	Order History 2017	Retention
Olist Brazilian E-Commerce	Order/Logistics Records 2016-2018	Fulfilment

34M+	3	10	6	15
Total Rows	Datasets	Phases	Statistical Tests	Interactive Charts

GitHub: github.com/NPK05/funnel-analysis-project | March 2026

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1 Technical Methodology

Ten-phase analytical workflow from problem definition to portfolio packaging

This project follows a structured ten-phase workflow designed to mirror professional data analysis practice. Every phase build on the previous analysis only proceeds after the prior phase is complete and documented.

1.1 Ten-Phase Workflow

Phase	Name	Description
1	Problem Definition	Hypotheses, KPIs, analytical questions defined for all 3 datasets
2	Environment Setup	Python venv, folder structure, Git/GitHub connected
3	Data Cleaning	38 issues identified and resolved with documented decisions
4	Exploratory Analysis	16 charts built, 26 key patterns identified per dataset
5	Core Funnel Analysis	Conversion rates, cohort funnels, revenue leakage quantified
6	Statistical Testing	Chi-square, Mann-Whitney U, Spearman, Wilson CI applied
7	Visualization	15 interactive Plotly charts + Streamlit dashboard deployed
8	Business Insights	Prioritized recommendations with quantified revenue impact
9	Cross-dataset Comparison	Three funnel archetypes compared and contrasted
10	Portfolio Packaging	GitHub README, report, live dashboard published

1.2 Tech Stack

Component	Tool / Version
Language	Python 3.11
Data Manipulation	pandas 2.3.3, numpy 2.4.3
Statistical Analysis	scipy 1.17.1 (chi-square, Mann-Whitney U, Spearman, Wilson CI)
Visualisation (EDA)	matplotlib, seaborn
Visualisation (Portfolio)	plotly 5.x - interactive HTML charts
Dashboard	Streamlit - deployed on Streamlit Community Cloud
Version Control	Git + GitHub
Environment	Python virtual environment (venv)

1.3 Statistical Tests Applied

Test	When used	Applied To
Chi-square	Comparing conversion rates between two groups	Mobile vs desktop CVR, country CVR, category CVR, monthly CVR, cadence loyalty, state late rate
Mann-Whitney U	Comparing non-normal distributions	Late vs on-time review scores (ordinal 1-5 scale), weekly vs monthly order counts
Spearman Correlation	Association between two continuous/ordinal variables	Delivery delay days vs review score
Wilson Score CI	Confidence intervals on proportions	All conversion rates, loyalty rates, late delivery rates

2 Dataset Descriptions and Cleaning

38 data quality issues identified and resolved across 34M+ rows

2.1 Dataset Overview

Dataset	Funnel Type	Volume	Users	Period	Output
Google Merch Store	Conversion	719,386 events	14,701 users	Nov 2020 - Jan 2021	25 cols
Instacart Grocery	Retention	33,819,106 rows	206,209 users	Historical (2017)	19 cols
Olist E-Commerce	Fulfilment	99,441 orders	99,441 customers	Sep 2016 - Oct 2018	43 cols

2.2 Professional Cleaning Philosophy

Every null value was categorized as one of three types before treatment:

- Random missing - tracking gap, no pattern. Treatment: flag with 'Unknown' or domain-appropriate fill.
- Structural missing - missing by design. Treatment: fill with logical value (e.g. 0 for first-order days_since_prior) + flag.
- Not-random missing - missingness IS the signal. Treatment: keep as null and document why.

> *Core principle: Never fill blindly. Every cleaning decision is documented with a reason. A null review score on a delivered order is not missing data - it is a non-response signal that tells us something about that customer.*

2.3 Key Cleaning Decisions

Dataset	Issue	Treatment	Reason
Google Merch	39,498 duplicate rows	Dropped	GA4 pipeline fired entire cart batch twice - confirmed by event_type distribution
Google Merch	408 null variants (29.5%)	Context-aware fill	Apparel: 'Size Not Specified'; others: 'Single Option Only'
Google Merch	2,902 session integrity flags	Flagged, kept	Purchase with no checkout event - GA4 session gap, not fake data
Instacart	206,209 null days_since_prior	Fill 0 + flag	All on order_number=1 - structural, first orders have no prior
Instacart	75,000 eval_set=test orders	Dropped	No product details available - unusable for product-level analysis
Instacart	Dataset constraint	Documented	Only users with 3+ orders included - order 1 to 2 drop-off unmeasurable
Olist	Timestamp nulls (2,965)	Kept as null	Nulls represent orders that never reached that stage - the funnel drop-off IS the null
Olist	551 duplicate reviews	Kept most recent	Customer updated their review - latest is most relevant
Olist	768 null review scores	Kept as null	Non-response bias - a delivered order with no review is a meaningful signal
Olist	1M geolocation rows	Aggregated to 19K	Prevent 52x row explosion on join - median lat/lng per zip code

3 Key Findings and Insights

Proven findings from Phase 5 EDA and Phase 6 statistical testing

3.1 Google Merchandise Store - Conversion Funnel

32.4%	49%	$552K	$65.12
Overall CVR ATC to Purchase	Primary Drop-off ATC to Checkout	Revenue Leakage 3-month window	Avg Order Value Per purchasing session

Stage	Users	Rate	Lost	Note
Add to Cart	12,545	85.3% of all users	-	-
Begin Checkout	6,404	51.0% of ATC users	6,141 lost	49.0% - LARGEST DROP
Purchase	4,066	63.5% of Checkout users	2,338 lost	36.5%
OVERALL CVR	-	32.4% (ATC to Purchase)	8,479 total lost	$552,179 leakage

Hypothesis disproved: Mobile (28.25%) converts nearly identically to desktop (27.28%). Chi-square p=0.208 - NOT statistically significant. The problem is device-agnostic checkout friction, not mobile UX.

Monthly cohort finding: November CVR 71.4% vs December 26.5% vs January 19.8%. All pairwise differences p<0.001. These are fundamentally different user cohorts - November buyers are high-intent gift purchasers, December brings holiday browsers.

Category finding: Campus Collection (3.6%), Office (3.4%), Drinkware (3.1%) all statistically outperform Apparel (1.83%) at p<0.001. Notebooks & Journals at 0.2% CVR is catastrophically low - 16,514 ATCs, only 36 purchases.

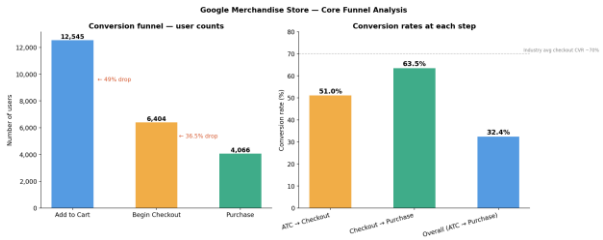

Fig 3.1a - Core Conversion Funnel

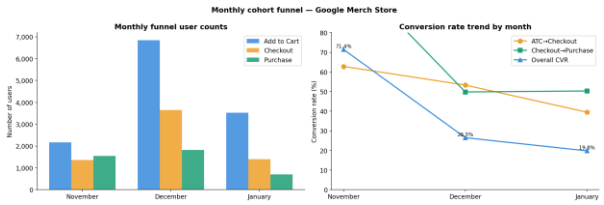

Fig 3.1b - Monthly Cohort Funnel

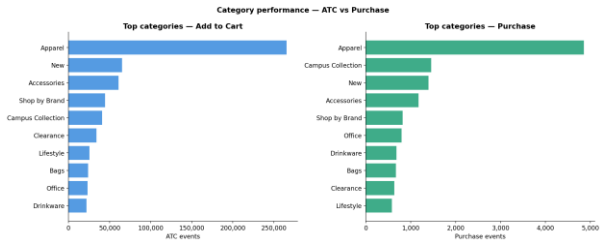

Fig 3.1c - Category ATC vs Purchase

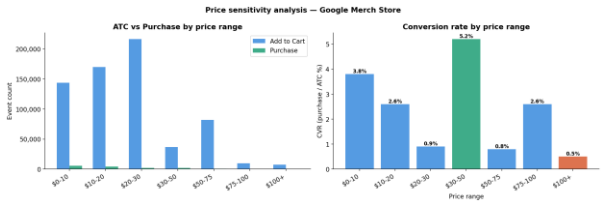

Fig 3.1d - Price Sensitivity Analysis

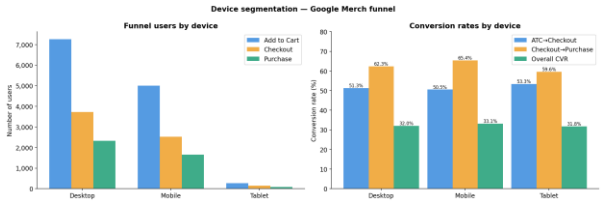

Fig 3.1e - Device Funnel (Mobile vs Desktop)

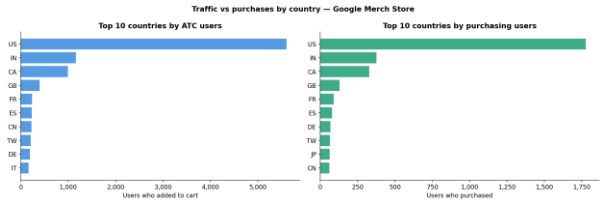

Fig 3.1f - Country Traffic vs Purchase

3.2 Instacart - Retention Funnel

76.0%	49,404	21.7pp	5x
Loyalty Rate 6+ orders CI [75.9%, 76.2%]	Users at Risk Stop at 3-5 orders (24%)	Cadence Gap Weekly 81% vs Monthly 59.3%	LTV Ratio Weekly vs Monthly users

Stage	Users	Rate	Note
First order	206,209	100%	Dataset entry - all users have 3+ orders
Second order	206,209	100%	Dataset constraint - filtered by Instacart
Active (3-5 orders)	206,209	100%	Dataset entry point for this analysis

Loyal (6+ orders)	156,805	76.0%	Primary measurable drop-off: 49,404 users stop

Strongest findings in project: Weekly ordering cadence users remain at 81.0% vs monthly users at 59.3% - a 21.7pp gap. Chi-square statistics = 2,731, p=0.000000. Weekly users also place 33.8 orders on average vs 6.7 for monthly - a 5x lifetime value difference (Mann-Whitney p=0.000000).

Critical intervention point: The steepest drop in the retention curve is between order 4 and order 5 - losing 10.9% of users at a single step. This is 2x the average drop rate. A targeted nudge (discount, push notification) after order 4 is the highest-ROI retention action.

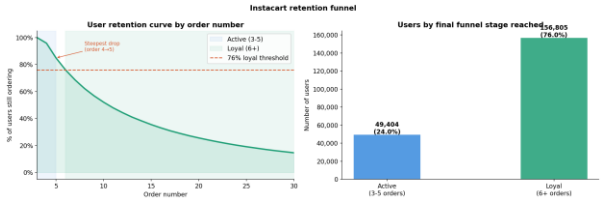

Fig 3.2a - Retention Funnel Stages

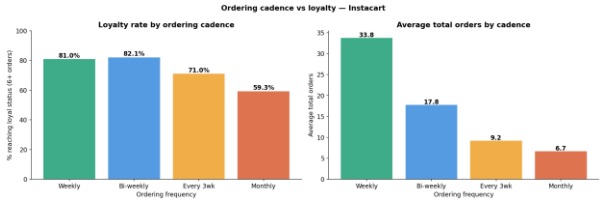

Fig 3.2b - Cadence Retention Rate

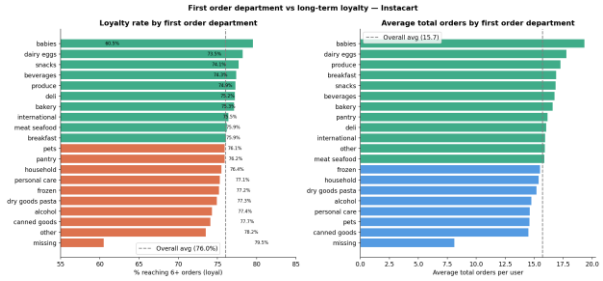

Fig 3.2c - Department Retention Analysis

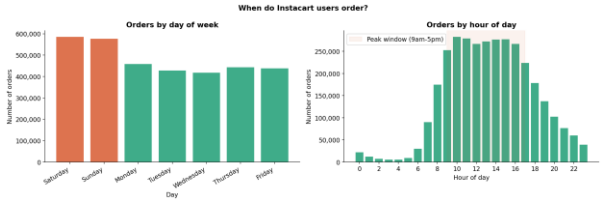

Fig 3.2d - Order Timing Patterns (Saturday 10am peak)

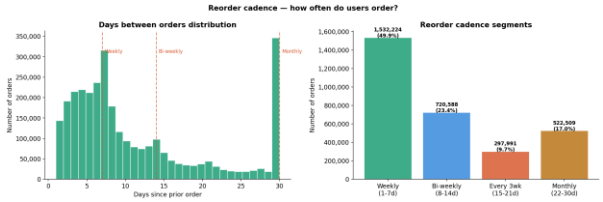

Fig 3.2e - Reorder Cadence Distribution

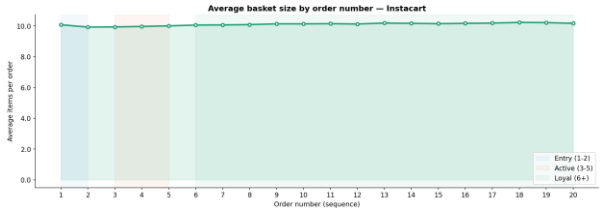

Fig 3.2f - Basket Size Trend

3.3 Olist Brazilian E-Commerce - Fulfilment Funnel

97.0%	**8.11%**	**1.73pts**	**R$1.85M**
Fulfilment Rate CI [96.9%, 97.1%	Late Rate 7,822 orders CI [7.94% 8.28%]	Score Gap Late 2.56 vs On-time 4.29	Revenue at Risk Direct + reputation impact

Stage	Orders	Rate	Dropped	Note
Created	99,441	100.0%	0 (0.0%)	-
Approved	99,281	99.8%	160 (0.2%)	Payment failures

Shipped	97,658	98.2%	1,623 (1.6%)	Seller fulfilment failures - LARGEST DROP
Delivered	96,476	97.0%	1,182 (1.2%)	Last mile failures
Reviewed	95,832	96.4%	644 (0.6%)	99.3% review rate among delivered orders

Non-linear threshold effect: The relationship between delivery delay and review score is not linear. Spearman r=-0.176 understates the real impact. Crossing from 'early' to 'late' drops the median review score from 5 stars to 2 stars - a cliff, not a gradient. Once more than 1 week late, 68-69% of customers give 1 star.

Geographic gap proven: AL (23.9% late rate) has 4x the late rate of SP (5.9%). RJ (13.5%, 12,350 orders) is the highest problem state. MG (5.6%) is statistically indistinguishable from SP (p=0.254) - shares logistics.

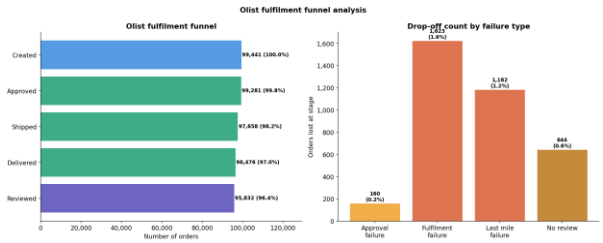

Fig 3.3a - Fulfilment Funnel Stages

Fig 3.3b - Delivery Delay vs Review Score

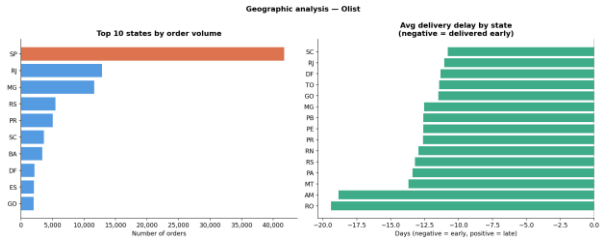

Fig 3.3c - Geographic Late Rate Analysis by State

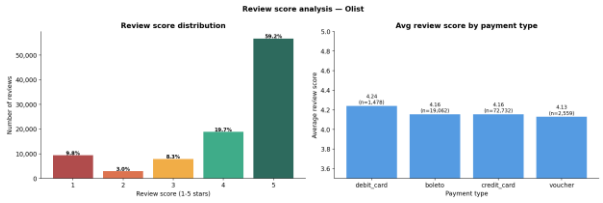

Fig 3.3d - Review Score Distribution

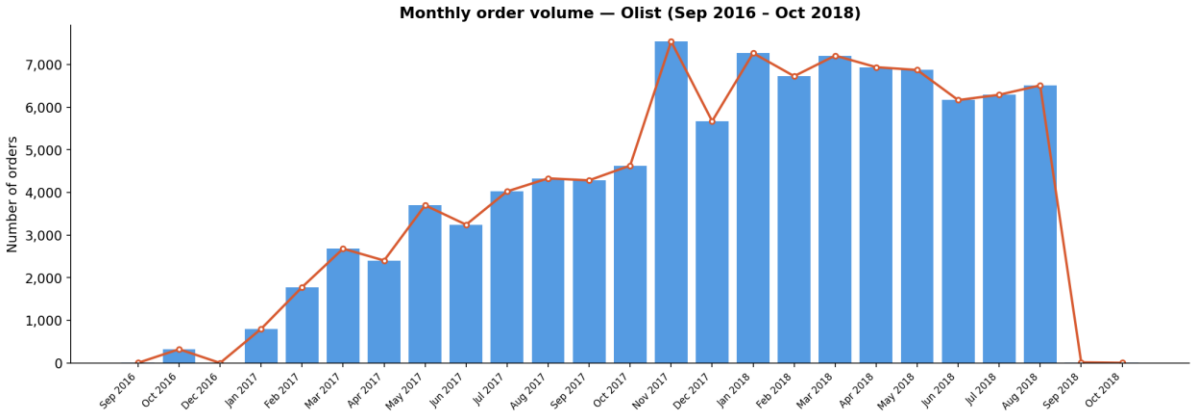

Fig 3.3e - Monthly Order Volume (Olist, Sep 2016 - Oct 2018)

4 Statistical Test Results

Every key finding validated with appropriate statistical test

Test	Dataset	Comparison	Values	Statistic	p-value	Conclusion
Chi-square	Google Merch	Mobile vs Desktop CVR	28.25% vs 27.28%	chi2=1.58	0.208	NOT SIGNIFICANT - Device not the problem
Chi-square	Google Merch	Campus vs Apparel CVR	3.57% vs 1.83%	chi2>>100	<0.001	SIGNIFICANT - Category differences real
Chi-square	Google Merch	Nov vs Dec CVR	71.4% vs 26.5%	chi2>>100	<0.001	SIGNIFICANT - Different user cohorts
Chi-square	Google Merch	FR vs US CVR	38.5% vs 31.7%	chi2=2.37	0.119	NOT SIGNIFICANT - Sample too small (n=234)
Chi-square	Instacart	Weekly vs Monthly loyalty	81.0% vs 59.3%	chi2=2,731	0.000000	HIGHLY SIGNIFICANT - Strongest finding
Chi-square	Instacart	Dept loyalty vs baseline	Babies 79.5% vs CG 74.1%	varied	<0.05	SIGNIFICANT - Weak effec (4.8pp)
Mann-Whitney U	Instacart	Weekly vs Monthly orders	Median 22 vs 6 orders	U=439M	0.000000	HIGHLY SIGNIFICANT - 5x LTV difference
Mann-Whitney U	Olist	Late vs On-time scores	2.56 vs 4.29 avg	U=524M	0.000000	HIGHLY SIGNIFICANT - 1.73pt gap proven
Spearman r	Olist	Delay days vs score	r=-0.176	-	0.000000	SIGNIFICANT but WEAK - Non-linear relationship
Wilson CI	Google Merch	Overall CVR	32.4%	CI [31.6%, 33.2%]	-	TIGHT CI - High precision
Wilson CI	Instacart	Loyalty rate	76.0%	CI [75.9%, 76.2%]	-	VERY TIGHT (0.37pp) - Most precise metric
Wilson CI	Olist	Late rate	8.11%	CI [7.94%, 8.28%]	-	TIGHT CI - Reliable measurement
Chi-square	Olist	AL late rate vs SP	23.9% vs 5.9%	chi2>>100	0.000000	SIGNIFICANT - 4x worse confirmed
Chi-square	Olist	MG late rate vs SP	5.6% vs 5.9%	chi2=1.30	0.254	NOT SIGNIFICANT - MG same as SP

Note on p=0.000000: This does not mean p is literally zero. Python rounds values smaller than 6 decimal places. Actual values may be 10^-47 or smaller. Always verify effect size alongside p-value - a large effect with p=0.000000 is valid; a tiny effect (from a massive sample) would be statistically real but practically meaningless.

5 Business Recommendations

Prioritised by revenue impact - proven findings only

5.1 Google Merchandise Store

#	Priority	Action	Est. Impact	Rationale
1	HIGH	Fix cart-to-checkout CTA friction	$46K/quarter (conservative)	6,141 users lost at ATC to Checkout. Industry avg 70%, curren 51%. Test: sticky CTA, exit-intent offers, reduce clicks to checkout.
2	HIGH	Investigate Notebooks & Journals 0.2% CVR	Unknown (potential quick win)	16,514 ATC events, only 36 purchases. Check for checkout bug, pricing mismatch, or product-market fit failure.
3	MED	Separate Nov/Dec campaign strategy	CVR improvement on existing traffic	November buyers are high-intent (71.4% CVR). December are browsers (26.5%). Treat as separate campaigns with different messaging.
4	MED	Promote $30-50 price range products	$20K+/quarter estimate	$30-50 converts at 5.2% - 3x the $20-30 range (0.9% CVR). Shift homepage merchandising toward this sweet spot.
5	LOW	Localization audit for CN and BE markets	Unknown	China (27.7%) and Belgium (16.4%) convert well below average. Check payment methods, currency display, shipping costs.

5.2 Instacart

#	Priority	Action	Scale	Rationale
1	HIGH	Intervene at order 4-5 with nudge or discount	~22,500 users per cohort	Steepest single drop in retention curve (10.9%). A targeted push notification or discount after order 4 is the highest-ROI action.
2	HIGH	Cadence acceleration: convert monthly to bi-weekly	17,673 users, 22pp loyalty improvement potential	Monthly loyalty 59.3% vs bi-weekly 82.1%. Tactic: 'weekly top-up' feature showing items running low based on purchase history.
3	MED	Protect weekly users with loyalty program	31,626 users, 5x LTV vs monthly	Weekly users place 33.8 orders on average - most valuable segment. Streak rewards, free delivery milestones, early access.
4	LOW	Surface babies/dairy/produce in onboarding	All new users	These departments drive highest loyalty (79.5%, 78.2%, 77.3%). Effect is real but weak (max 4.8pp). Prioritise cadence over department.

5.3 Olist Brazilian E-Commerce

#	Priority	Action	Impact	Rationale
1	HIGH	Fix RJ delivery reliability	~R$269K direct, ~1,670 late orders rescued	12,350 orders at 13.5% late rate - highest volume problem state. Negotiate priority carrier SLAs for Rio de Janeiro routes.
2	HIGH	Late delivery early warning system	Protect full R$1.35M	Flag orders at risk T-3 days before estimated delivery. Proactive R$20 voucher converts near-certain 1-star into loyalty opportunity.

3	MED	Buffer delivery estimates for northeast states	AL, MA, CE - ~2,590 orders	AL (23.9%), MA (19.7%), CE (15.3%) - structural logistics gap. AM state proves slow+reliable beats fast+unreliable.
4	MED	Seller accountability for fulfilment failures	1,623 orders approved but not shipped	Customers paid and received nothing. Implement seller reliability scoring - suspend sellers with >2% non-fulfilment rate
5	LOW	Replicate MG logistics model in other states	Medium-term infrastructure	MG (5.6% late rate) is statistically same as SP (p=0.254). Study what MG does differently and apply to BA, CE, SE.

6 Visualizations and Charts

15 interactive Plotly charts + Streamlit dashboard

All charts are interactive HTML files saved to outputs/dashboard/ and integrated into a Streamlit dashboard deployed on Streamlit Community Cloud. The dashboard features a dataset selector and tabbed navigation covering the full funnel picture for each dataset.

6.1 Chart Index

#	Dataset	Chart Title	Type	What It Shows
01	Google Merch	Core Conversion Funnel	Plotly Funnel	ATC to Checkout to Purchase with drop-off annotations
02	Google Merch	User Flow Sankey	Plotly Sankey	Full user journey showing all conversion and abandonment paths
03	Google Merch	Segmented Funnel	Grouped Bar	Device CVR comparison + Top 12 countries by CVR with US baseline
04	Google Merch	Monthly CVR Trend	Line + CI Band	Nov/Dec/Jan CVR with 95% confidence interval shading
05	Google Merch	Revenue Leakage Waterfall	Waterfall	Potential to Cart loss to Checkout loss to Captured
06	Instacart	Retention Funnel	Bar Chart	Funnel stages with dataset constraint documentation
07	Instacart	Retention Curve	Area Line	Decay curve by order number with steepest drop annotated
08	Instacart	Cadence Loyalty	Bar + Error Bars	Loyalty rate by cadence with 95% CI error bars
09	Instacart	Department Loyalty	Dual Horizontal Bar	Loyalty rate and avg orders side by side by first order department
10	Instacart	Order Timing Heatmap	Heatmap	Order volume by day x hour - Saturday 10am peak visible
11	Olist	Fulfilment Funnel	Plotly Funnel	5-stage funnel with seller failure annotation
12	Olist	Delivery vs Review Score	Dual Bar	Avg score and 5-star/1-star % by delivery timing bucket

13	Olist	Brazil State Performance	Bubble Map + Bar	Scattergeo bubble map + ranked bar charts
14	Olist	Revenue Impact Waterfall	Waterfall	Direct revenue + reputation loss + recovery scenario
15	All Datasets	Cross-dataset Comparison	Grouped Bar	Three funnel archetypes side by side on same scale

6.2 Dashboard Architecture

The Streamlit dashboard (scripts/dashboard.py) is a single-file app with three dataset views selected from the sidebar. Each view contains multiple tabs covering the full analytical story:

Dataset	Tabs
Google Merch	Funnel Overview | Device & Country | Category Analysis | Monthly Cohorts | Revenue Impact
Instacart	Retention Funnel | Ordering Cadence | Department Analysis | Order Timing
Olist	Fulfilment Funnel | Delivery vs Reviews | Geographic Analysis | Revenue Impact

6.3 Additional EDA Charts

Selected exploratory data analysis charts from all three datasets:

Fig 6.1 - Google Merch: Daily Event Volume

Fig 6.2 - Google Merch: Hourly Purchase Pattern

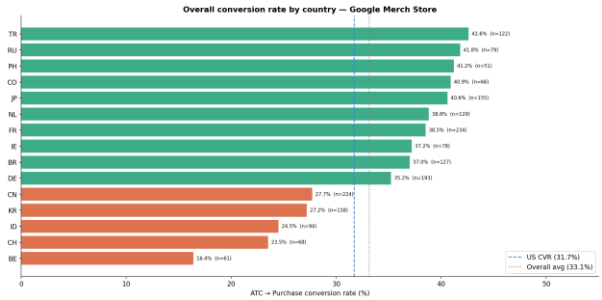

Fig 6.3 - Google Merch: Country Funnel

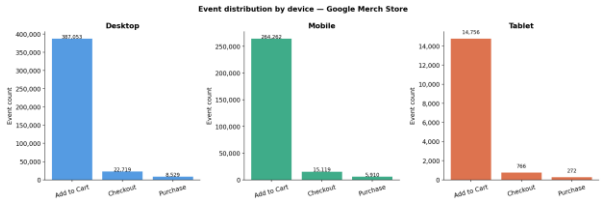

Fig 6.4 - Google Merch: Device Event Distribution

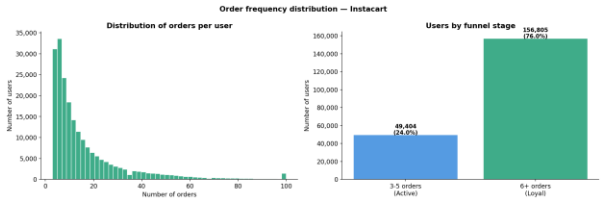

Fig 6.5 - Instacart: Order Frequency Distribution

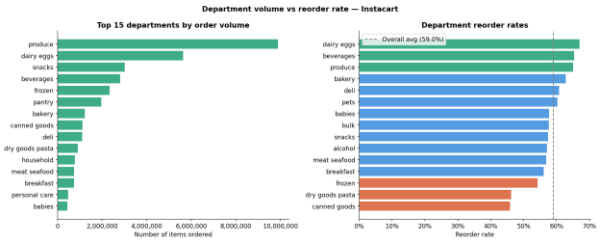

Fig 6.6 - Instacart: Department Analysis

7 Lessons Learned and Challenges

Technical problems solved and analytical insights gained

7.1 Technical Challenges

GA4 Pipeline Duplication (Google Merch)

Discovered 39,498 exact duplicate rows concentrated in begin_checkout events. Initially appeared as a data quality error. Root cause: GA4 fired the entire cart batch twice as a pipeline artifact. This would have made checkout CVR appear artificially low if not caught. Fix: drop_duplicates() after confirming identical timestamps and session IDs.

Instacart 33.8M Row Memory Management

Loading the full master file took 5 minutes and consumed 2GB+ RAM. Solution: dtype optimization (int8/int16/int32) saved 1,385MB. For EDA, used 10% random sample (skiprows lambda) which loaded in 20 seconds and gave statistically accurate distributions with 3.3M rows.

Olist Review Join Row Explosion

551 orders had multiple reviews, causing the master table to expand from 99,441 to 99,992 rows on join. Fix: deduplicate reviews by order_id keeping most recent (customer updated their review), then rejoin cleanly.

GitHub 100MB File Size Limit

Cleaned CSV files exceeded GitHub's 100MB limit. Fix: added data/cleaned/ to .gitignore, used git rm --cached to remove already-tracked files, then git filter-branch and force push to clean history.

Streamlit Cloud requirements.txt Conflict

pip freeze generated a 100+ package requirements.txt including Jupyter, notebook, and dev dependencies. Streamlit Cloud hit version conflicts. Fix: minimal requirements.txt with only 6 directly imported packages.

Brazil Choropleth Map GeoJSON Failure

Initial choropleth approach failed because the GeoJSON URL required internet access which was blocked locally. Solution: switched to Plotly Scattergeo bubble map using state centroid coordinates - no external dependency, works offline and on Streamlit Cloud.

7.2 Analytical Lessons

Hypotheses Exist to Be Disproved

The original hypothesis - mobile converts lower than desktop - was statistically disproved (p=0.208). This is a better outcome than confirmation. A disproved hypothesis redirects analytical focus to where the real problem actually is. The real problem was checkout initiation friction, not device UX.

Missingness is Data

Olist delivery timestamp nulls are not missing data - they are orders that never reached that stage. Olist review score nulls are not random gaps - they are non-response signals. Filling them would have destroyed analytical value. The question 'why is this null?' is always more important than 'how do I fill it?'

Statistical Significance vs Practical Significance

With 200K+ users, even tiny differences become statistically significant. Instacart department loyalty differences are statistically real ($p<0.05$) but practically weak (max 4.8pp). Cadence differences are both statistically certain ($p=0.000000$) and practically enormous (21.7pp). Always report effect size alongside p-value.

Non-linear Relationships Resist Linear Metrics

The Spearman correlation between Olist delivery delay and review score was $r=-0.176$ (weak). But the Mann-Whitney test showed a 1.73-point gap between late and on-time orders. The relationship is threshold-based: being late causes a cliff-drop, not a gradient. Linear correlation understated the real business impact significantly.

Product Type Determines Behaviour Pattern

Google Merch peaks on Wednesday (office workers browsing branded merchandise). Instacart peaks on Saturday 10am (weekly grocery planning). The same analytical framework applied to different products produces completely different insights. Never assume findings from one domain transfer to another.

Sample Size Determines Answerable Questions

Google Merch country differences looked large (France 38.5% vs US 31.7%) but were not statistically significant ($p=0.119$) because France had only 234 ATC users. Instacart cadence differences were instantly significant with 17,000+ users per segment. Sample size is not just data quality - it determines which hypotheses are testable.

8 Project Summary

Cross-dataset comparison and key deliverables

8.1 Cross-Dataset Comparison

Metric	Google Merch	Instacart	Olist
Funnel type	Conversion	Retention	Fulfilment
Completion rate	32.4%	76.0%	97.0%
Biggest single drop	ATC to Checkout (49%)	Active to Loyal (24%)	Approved to Shipped (1.6%)
Root cause	User decision (UX/price friction)	Habit failure (cadence not formed)	Operational failure (seller/logistics)
Right intervention	Reduce checkout friction	Nudge at order 4-5 to build habi	Accountability + infrastructure
Revenue impact	$552K leakage (3 months)	LTV gap: 5x weekly vs monthly	R$1.85M at risk from late orders
Peak behaviour	Wednesday office hours	Saturday 10am grocery	November Black Friday BR
Sample size	14.7K users, 719K events	206K users, 33.8M rows	99K orders, 9 tables

Core cross-dataset insight: The right metric and the right intervention are completely determined by the funnel archetype. Applying conversion rate analysis to a fulfilment funnel, or retention analysis to a conversion funnel, would produce misleading conclusions. Understanding which type of funnel, you are analysing is the most important analytical decision in the entire project.

8.2 Project Deliverables

Deliverable	Description	Location / URL
GitHub Repository	Public repo with all notebooks, scripts, and outputs	github.com/NPK05/funnel-analysis-project
Jupyter Notebooks	4 notebooks covering all 10 phases	01_google_merch. ipynb, 02_instacart.ipynb, 03_olist.ipynb, 04_comparison.ipynb
Cleaned Data	6 production-ready CSV files (excluded from Git - reproducible from notebooks)	data/cleaned/ - 34M+ rows across 3 datasets
EDA Charts	16 matplotlib/seaborn charts for exploratory analysis	outputs/charts/google/, instacart/, olist/
Portfolio Charts	15 interactive Plotly HTML charts	outputs/dashboard/01_google_funnel.html through 15_cross_dataset_comparison.html
Streamlit Dashboard	Live interactive dashboard with dataset selector + tabbed navigation	scripts/dashboard.py - deployed on Streamlit Community Cloud
Portfolio Report	This document - full project documentation	Funnel Analysis Portfolio Report.docx